SPHOORTHY ENGINEERING COLLEGE

An Autonomous Institution Approved by AICTE & Accredited with NAAC Grade 'A'
Department of Examination & Academic Affairs · Hyderabad — 501510

Database Management Systems: Relational Models & SQL

Program / Dept: B.Tech / MBA — CSE	**Course Code / Title:** Database Management Systems (CS402PC)
Academic Regulation: Autonomous (SR23 / SR24)	**Curriculum Semester:** Semester 4 (Year 2)
Credits / Contact Hours: 3 Credits · 48 Lecture Hours	**Evaluation Scheme:** Continuous Internal (40M) + End Sem (60M)

Course Overview & Pedagogical Objectives:

Entity-Relationship Diagrams, Relational Algebra, SQL queries, Normalization (1NF through BCNF), Transaction Processing (ACID), and Concurrency Control. The primary objective of this course is to equip students with sound theoretical rigor, analytical problem-solving competencies, and real-world system engineering capabilities demanded by premier technological organizations and competitive graduate benchmarks.

Course Outcomes (COs) & Bloom's Taxonomy Mapping:

CO ID	Course Learning Outcome Description	Bloom's Level
CO 1	Understand and formulate fundamental concepts of Unit 1: Database Architecture & ER Model.	L2 Understand
CO 2	Analyze algorithms, mechanisms, and models in Unit 2: Relational Algebra & Calculus.	L4 Analyze
CO 3	Design and evaluate robust architectures for Unit 3: SQL & Database Normalization.	L5 Evaluate
CO 4	Implement and optimize operational pipelines in Unit 4: Transaction & Concurrency Control.	L3 Apply
CO 5	Integrate state-of-the-art frameworks in Unit 5: Storage and Indexing for industrial solutions.	L6 Create

Comprehensive Module Index:

Unit	Title & Core Concepts	Page
Unit 1	**Unit 1: Database Architecture & ER Model**: 3-tier database architecture, Data independence, Strong & weak entity sets, Cardinality constraints, and Exten...	Page 2
Unit 2	**Unit 2: Relational Algebra & Calculus**: Fundamental operators (Select, Project, Cartesian Product, Union, Set Difference), Rename, Joins (Theta, Equi,...	Page 3
Unit 3	**Unit 3: SQL & Database Normalization**: DDL, DML, DCL, Aggregate functions, Nested subqueries, Functional dependencies, Armstrong axioms, 1NF, 2NF, 3N...	Page 4
Unit 4	**Unit 4: Transaction & Concurrency Control**: ACID properties, Serializability (Conflict & View), Two-Phase Locking (2PL), Deadlock prevention and detection...	Page 5
Unit 5	**Unit 5: Storage and Indexing**: RAID levels, B-Trees and B+ Trees indexing structure, Static and Dynamic hashing, and Query optimization execu...	Page 6

UNIT 1: DATABASE ARCHITECTURE & ER MODEL • *Core Theory & Practice Guide*

1.1 Technical Exposition & Conceptual Foundations

The study of unit 1: database architecture & er model forms an indispensable cornerstone of Database Management Systems (CS402PC). Specifically, 3-tier database architecture, Data independence, Strong & weak entity sets, Cardinality constraints, and Extended ER features. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

1.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 1 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

1.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 1
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

1.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 1 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 1. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 1 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 1 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 2: RELATIONAL ALGEBRA & CALCULUS • *Core Theory & Practice Guide*

2.1 Technical Exposition & Conceptual Foundations

The study of unit 2: relational algebra & calculus forms an indispensable cornerstone of Database Management Systems (CS402PC). Specifically, Fundamental operators (Select, Project, Cartesian Product, Union, Set Difference), Rename, Joins (Theta, Equi, Natural, Outer), and Tuple Relational Calculus. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

2.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 2 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

2.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 2
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

2.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 2 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 2. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 2 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 2 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 3: SQL & DATABASE NORMALIZATION • *Core Theory & Practice Guide*

3.1 Technical Exposition & Conceptual Foundations

The study of unit 3: sql & database normalization forms an indispensable cornerstone of Database Management Systems (CS402PC). Specifically, DDL, DML, DCL, Aggregate functions, Nested subqueries, Functional dependencies, Armstrong axioms, 1NF, 2NF, 3NF, and Boyce-Codd Normal Form (BCNF). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

3.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 3 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

3.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 3
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

3.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 3 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 3. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 3 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 3 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 4: TRANSACTION & CONCURRENCY CONTROL • *Core Theory & Practice Guide*

4.1 Technical Exposition & Conceptual Foundations

The study of unit 4: transaction & concurrency control forms an indispensable cornerstone of Database Management Systems (CS402PC). Specifically, ACID properties, Serializability (Conflict & View), Two-Phase Locking (2PL), Deadlock prevention and detection, and Timestamp-based protocols. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

4.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 4 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

4.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 4
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

4.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 4 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 4. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 4 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 4 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 5: STORAGE AND INDEXING • *Core Theory & Practice Guide*

5.1 Technical Exposition & Conceptual Foundations

The study of unit 5: storage and indexing forms an indispensable cornerstone of Database Management Systems (CS402PC). Specifically, RAID levels, B-Trees and B+ Trees indexing structure, Static and Dynamic hashing, and Query optimization execution plans. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

5.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 5 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

5.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 5
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

5.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 5 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 5. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 5 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 5 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

Standard Recommended Textbooks & Digital References:

1. *Primary Curriculum Standard Textbook* — Pearson Higher Education / McGraw-Hill Education (Latest Edition).
2. *Autonomous Reference Syllabus Blueprint* — Sphoorthy Engineering College Academic Council.

3. *NPTEL & IEEE Computer Society Guidelines* — Open Educational Resources & Peer-reviewed Publications.